

From Aggregate Scores to Warranted Inference in AGI Evaluation

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Abstract

Finite multidomain batteries are used to support claims about untested tasks, contexts, systems, and future behaviour. Yet aggregate stability can conceal offsetting item-level changes, while a change-based tail statistic can miss failures that are severe in both tested conditions. I use PROJECTIBILITY for target-indexed support for a specified inference beyond the observations used to construct a score. Projectibility belongs within, rather than in place of, a broader validity argument.

I develop an evaluation discipline with three moves. First, declare the projection target, system and task populations, bearer, intervention range, time horizon, decision loss, and evidential test. Second, preserve five distinct descriptions: domain-profile correlation, signed level change, item instability, worst-tail degradation, and absolute worst-tail loss. Third, test the projection using holdouts at the relevant facet; causal and decision claims require interventions or loss-sensitive evidence.

Interpretable aggregates are separated from predictive or decision models, while retention, feedback, and component dependence are treated as applications with their own controls. A reanalysis of 32 released model–benchmark cells and known-truth simulations shows what the granular descriptions distinguish and why pseudo-null-adjusted and split-tail estimates aren’t interchangeable. The result is a target-indexed evaluation framework, not a new general-purpose score.

I THE INFERENTIAL OVERREACH OF AGI SCORES

An unchanged average can conceal a changed system.

The argument has three moves. First, declare the inference or decision and the scope over which it is meant to hold. Second, preserve distinctions that aggregation compresses: domain-profile correlation, signed level, two-sided item change, change in the worst tail, and absolute tail loss. Third, test the proposed inference with evidence not used to construct the score and matched to the declared scope. Granular description preserves candidate evidence; it doesn’t, by itself, warrant a broader claim.

Zhang et al. (2026) add task-irrelevant context to benchmark items and find that aggregate accuracy often moves little even when many item-level response probabilities move in opposite directions.

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Their adjusted estimates reach 13.6% for mean absolute item change and 53.2% for worst-decile degradation across the reported model–benchmark cells. The affected examples are also largely model-specific.

The immediate lesson isn’t that aggregate accuracy is useless. A stable mean answers only a question about the mean. The same value is compatible with no item changes, balanced improvements and degradations, or a stable set of serious failures. Those states may support different predictions and decisions.

The difficulty becomes acute when a finite battery is interpreted as evidence of general capability. Raji et al. (2021) argue that broad benchmark claims routinely outrun the contextual tasks used to construct them. Hendrycks et al. (2025), for example, organize AGI evaluation as a ten-domain battery grounded in Cattell–Horn–Carroll (CHC) theory. The move from isolated leaderboards to a multidomain profile is valuable, but breadth within a battery doesn’t establish which inferences the profile supports beyond it.

Hendrycks et al.’s battery is one motivating case of a general measurement problem. When an AGI score is used to support claims about novel items, contexts, systems, or future behaviour, its intended range is exceptionally broad relative to a finite construction sample. The same framework applies to multidomain AI evaluation more generally, but AGI claims make the gap especially visible.

The contribution is a target-indexed evaluation discipline rather than a novel family of mathematical operations. Its three moves are to declare the projection before choosing the score, preserve the relevant levels of description, and test the projection against a simple baseline on matching evidence. Aggregation is warranted only when held-out evidence for a shared projection target supports it. Retention, feedback response, and component dependence later illustrate how the same discipline changes when the time horizon, intervention, or bearer changes.

2 FROM OBSERVATIONS TO DECLARED PROJECTIONS

Validity attaches to proposed interpretations and uses of scores, not to a test in isolation (Messick, 1989, 1995). I use PROJECTIBILITY, following the philosophical vocabulary associated with warranted projection (Boyd, 1991; Goodman, 1955), for one question within the broader validity argument: does the evidence support a specified inference beyond the observations used to construct the score? Projectibility doesn’t replace construct validity, generalizability, transportability, or decision analysis.

Different inferences and uses require different warrants. Table 1 separates six relations that a broad phrase such as “the score generalizes” can obscure. A decision isn’t itself an epistemic projection. It depends on projectible estimates together with a declared loss, utility, or constraint, and is included because score use belongs to the wider validity argument.

For causal effects, this point has a precise analogue: transporting an effect between populations requires assumptions about which causal mechanisms change and which remain invariant, not merely good held-out prediction (Pearl & Bareinboim, 2014). The present framework is mostly predictive and decision-oriented. It invokes causal transport only to mark a stronger claim that ordinary benchmark validation doesn’t establish.

Before calculation, an evaluator should register a PROJECTION DECLARATION:

1. Claim and projection target: the outcome, interpretation, or explanation to be inferred
2. System population: the model families, versions, configurations, settings, and users in scope
3. Task population: the item or task universe and its sampling frame

Table 1: Inferential relations, decision use, and evidence that could warrant them.

Claim or use	Declared range	Matching evidence
Tested to untested items	Item or task population	Item- or template-cluster holdout from a stated sampling frame
One context to another	Realizations or perturbation families	Hold out realizations, or entire families for the stronger claim
One system population to another	Models, lineages, or configurations	Hold out the corresponding systems rather than near-identical versions
Present to future behaviour	Sessions, releases, or deployment periods	Chronological validation at the stated horizon
Behaviour to causal explanation	Mechanisms and rival routes	Mechanistic intervention and controls for alternative causes
Score to decision	Actions and consequences	Independent calibration plus a declared loss, utility, or constraint

4. Bearer and unit: a base model, stateful agent, or deployed model–tool–retrieval composite
5. Intervention range: the perturbation families and realizations over which the claim is meant to hold
6. Time horizon: immediate, delayed, or later deployment behaviour
7. Decision criterion: the loss, utility, constraint, or action threshold when a decision is intended
8. Evidential test: the holdout unit or other evidence that could support or defeat the claim

A separate sampling audit records what generated the observations: model and version, item, domain, perturbation family and realization, response replicate, scorer, and time. These facets may be crossed, nested, fixed, or sampled. Generalizability theory provides the established vocabulary of an admissible universe, multiple facets, and a decision-specific measurement design (Brennan, 2001). This proposal adds signed, paired, and tail-sensitive descriptions; it doesn’t replace that machinery.

Here and below, a system “lineage” is an operational dependency group defined by declared provenance, training, distillation, shared components, or data dependence, not merely by a family label.

Consider a running release decision. A committee wants to predict serious task loss for a deployed agent when it encounters a previously unseen family of irrelevant contexts over the next release cycle. The system population is the product lineage, the item population is the declared deployment-task frame, the bearer is the full agent, the outer holdout is a later version crossed with a held-out context family, and the release rule includes an absolute tail-loss constraint. A random item split within familiar contexts wouldn’t test that projection.

Field interests determine which target matters. They don’t determine whether the evidence supports it. Projectibility is target-indexed without becoming conventional: the target is chosen, but success is constrained by the appropriate warrant, whether matched holdout performance, calibration and decision value, or intervention against rival explanations.

3 WHAT AGGREGATE STABILITY DOESN'T ESTABLISH

Aggregation can collapse distinctions at several levels. A stable mean within a domain can't distinguish unchanged items from item-level churn. A stable ten-domain profile can't distinguish a broadly preserved pattern from a narrow construction or subprocess that fails sharply. A stable average across models can't distinguish shared from family-specific failure sets.

Analogous cases arise outside broad capability testing. von der Malsburg and Padó (2026) find that transformers can appear human-aligned for one agreement-attraction configuration while diverging in another, directly warning against projection from an isolated configuration or model. Kuribayashi et al. (2026) find that different layer-specific surprisal measures align best with different aspects of human sentence processing. The latter result supports target-dependent measure selection rather than item-level disaggregation. Together, the cases show why a broad label doesn't determine which description supports a particular inference.

3.1 DATA STRUCTURE AND THREE ESTIMAND MODES

A full observation may be indexed by system and version, domain, item, perturbation family, perturbation realization, response replicate, scorer, and time. The facets can be crossed, nested, fixed, or sampled. Counting output rows as independent observations would generally be wrong.

To keep the definitions readable, fix a system, time, scorer protocol, and perturbation realization. Let domains be indexed by $g = 1, \dots, G$, items within domain g by $i = 1, \dots, N_g$, and conditions by c , with baseline 0 and perturbation p . Let $s_{igc} \in [0, 1]$ be the item score. Define

$$\delta_{igp} = s_{igp} - s_{ig0}, \quad a_{gc} = \frac{1}{N_g} \sum_{i=1}^{N_g} s_{igc},$$

and let $\mathbf{a}_c = (a_{1c}, \dots, a_{Gc})$.

The same symbols support three different estimands. With deterministic decoding on a fixed paired set, the quantities below are exact descriptions of that set. Under a declared stochastic generation and scoring protocol, s_{igc} is an expected score estimated from repeated responses and, when relevant, scorers. Inference to an item, context, system, or time population also requires sampling or holdout from that declared universe. Repetition within an item doesn't create a representative item sample.

If item inclusion probabilities differ, population means require design weights and a q -tail should represent q of the target-population mass; the unweighted formulas below describe a fixed set or equal-probability sample.

Table 2 gives the minimal descriptive set. The raw domain profiles and item-change distributions remain part of the report; the displayed quantities don't replace them.

3.2 PROFILE CORRELATION AND SIGNED LEVEL

For nonconstant profiles, report Pearson correlation directly:

$$r_p = \text{corr}_g(\mathbf{a}_0, \mathbf{a}_p) \in [-1, 1].$$

The unit-interval transform $(1 + r_p)/2$ adds no information and invites ratio-like interpretations, so it isn't used. A high r_p establishes only strong linear association between the centred, scaled profiles. Any positive affine transformation of the profile can yield $r_p = 1$, even when level or spread changes.

Table 2: Distinct descriptions preserved before target-specific validation.

Output	Granularity	Question answered
r_p	Domain profile	How strongly were the centred, scaled domain profiles linearly associated?
L_{gp}, L_p	Domain and overall level	Did average performance rise or fall?
INS_{gp}	Items within a cell	How much two-sided paired item change occurred?
$WTD_{q,gp}$	Change tail	How large was mean degradation in the worst-changing q -fraction?
$WTL_{q,gc}^{(T)}$	Absolute loss tail	How large was target-specific loss in the worst-loss q -fraction?

With ten fixed domains, r_p is a descriptive summary of those ten estimated means. Resampling items within domains propagates uncertainty in the means, but not uncertainty in the choice of domains. Correlation is undefined when either profile has zero variance and degenerates to ± 1 with only two nonconstant domains. Reports should publish the profiles and their spreads; a preregistered centred distance or rank correlation may supplement r_p when it better matches the target.

Signed domain and overall changes are

$$L_{gp} = a_{gp} - a_{g0}, \quad L_p = \frac{1}{G} \sum_{g=1}^G L_{gp}.$$

They lie in $[-1, 1]$ for unit-interval scores. The equal-domain mean is a transparent default description, not a measurement-neutral claim that all domains have equal interval meaning. An item-pooled, unequally weighted, or deployment-mixture mean has to be labelled and justified.

3.3 ITEM INSTABILITY AND DIRECTIONAL COMPONENTS

Signed change can cancel within a domain. Mean absolute paired change is

$$INS_{gp} = \frac{1}{N_g} \sum_{i=1}^{N_g} |\delta_{igp}| \in [0, 1]. \quad (1)$$

Its directional components are

$$F_{gp}^+ = \frac{1}{N_g} \sum_i (\delta_{igp})_+, \quad F_{gp}^- = \frac{1}{N_g} \sum_i (-\delta_{igp})_+.$$

Here F^+ is mean positive change and F^- is mean negative change, recorded as a nonnegative magnitude. For deterministic binary scores, these are the incorrect-to-correct and correct-to-incorrect transition proportions. The identities

$$L_{gp} = F_{gp}^+ - F_{gp}^-, \quad INS_{gp} = F_{gp}^+ + F_{gp}^-$$

make cancellation explicit and imply $|L_{gp}| \leq INS_{gp}$.

3.4 WORST-TAIL DEGRADATION IS A CHANGE STATISTIC

Choose $q \in (0, 1]$ and a minimum tail count before inspecting results. Let $m_g = \lceil qN_g \rceil$, and order paired changes from smallest to largest. Define

$$\text{WTD}_{q,gp} = -\frac{1}{m_g} \sum_{\ell=1}^{m_g} \delta_{(\ell)gp} \in [-1, 1]. \quad (2)$$

WTD is a finite-sample expected-shortfall analogue applied to the loss $-\delta$ (Acerbi & Tasche, 2002; Rockafellar & Uryasev, 2002). Because the ceiling makes the empirical tail mass m_g/N_g , the selected set is exactly a q -tail only when those quantities coincide.

A positive WTD indicates mean deterioration among the worst-changing items. A negative value means that even the selected lower tail improved on average. The statistic measures signed worst-tail change, not absolute harm. Calling it a harmful-tail measure would import consequences that correctness change alone doesn't encode.

3.5 ABSOLUTE WORST-TAIL LOSS

A change statistic misses a stable serious failure. If the same safety-critical items are failed at baseline and under perturbation, their δ_{igp} values are zero, so both INS and WTD can be zero. The running release decision still needs an absolute loss constraint.

For target T , declare an item loss $\lambda_{igc}^{(T)} \in [0, 1]$ that incorporates the response, reference outcome, and any consequence severity relevant to the decision. Order condition-specific losses from largest to smallest and define

$$\text{WTL}_{q,gc}^{(T)} = \frac{1}{m_g} \sum_{\ell=1}^{m_g} \lambda_{[\ell]gc}^{(T)} \in [0, 1]. \quad (3)$$

Report $\text{WTL}_{q,g0}^{(T)}$ and $\text{WTL}_{q,gp}^{(T)}$ separately. Their difference can be secondary, but it can't replace their levels. When consequences differ across items, a binary 1 – correctness loss is inadequate; severity belongs in the declared target loss. If loss isn't normalized, report it in its natural units instead of claiming the unit-interval bound.

3.6 ESTIMATION, SELECTION, AND MULTIPLICITY

With stochastic generation or scoring, absolute values give the statistic a positive expectation under the response pseudo-null, while selecting the worst observed tail selects noise. Balanced repeated samples can estimate conditional response probabilities. Following Zhang et al. (2026), baseline-only pseudo-null contrasts should reproduce the real response counts. Reports should show the raw statistic, mean pseudo-null contribution, untruncated adjusted value, and uncertainty. This correction isn't an unbiased-recovery guarantee.

Tail selection and estimation should use disjoint response splits when item changes or losses are noisy. Cross-fitting addresses reuse of response noise; it doesn't guarantee recovery of the latent population tail. Deterministic fixed-set descriptions have no response-sampling contribution to remove, although uncertainty about new items or contexts remains.

Resampling and holdout units should follow the projection declaration. Preserve paired items, shared baselines, perturbation realizations, scorers, and model lineage. An item bootstrap conditional

on fixed domains and perturbations doesn't support claims about new domains or perturbation families. When a descriptive report contains many cells or tail fractions, distinguish cellwise from simultaneous uncertainty and state how selection or multiplicity is controlled.

These estimands presuppose a sound benchmark. Disaggregation can't resolve ambiguous items or an ill-designed perturbation. Annotation quality, statistical power, construct coverage, and the invariant property under stress testing have to be justified independently (Bowman & Dahl, 2021).

3.7 WHY NO GENERAL ROBUSTNESS SCALAR FOLLOWS

The quantities above aren't noisy indicators of one latent robustness variable. A high r_p with negative L_p means that the centred, scaled profiles remained linearly associated while level fell. Near-zero L_{gp} with high INS means that changes cancelled. High WTD localizes degradation in a change tail, while high WTL can reveal stable absolute loss even when every change statistic is zero.

The primary output is a multilevel descriptive report indexed by system, domain, and perturbation. A scalar is warranted only when held-out performance for a named projection target supports a function of these descriptions. The result is then a target-specific predictive or decision model, not a universal index of general intelligence.

4 WORKED DEMONSTRATION

The framework needs more than algebraic examples. The **empirical companion** uses the released trial-level outputs from Zhang et al. (2026) and known-truth simulations. It has three aims: reproduce the distinction between aggregate change and paired item instability; compare raw, pseudo-null-adjusted, and split-sample tail estimators; and demonstrate why a change tail can't replace an absolute loss tail.

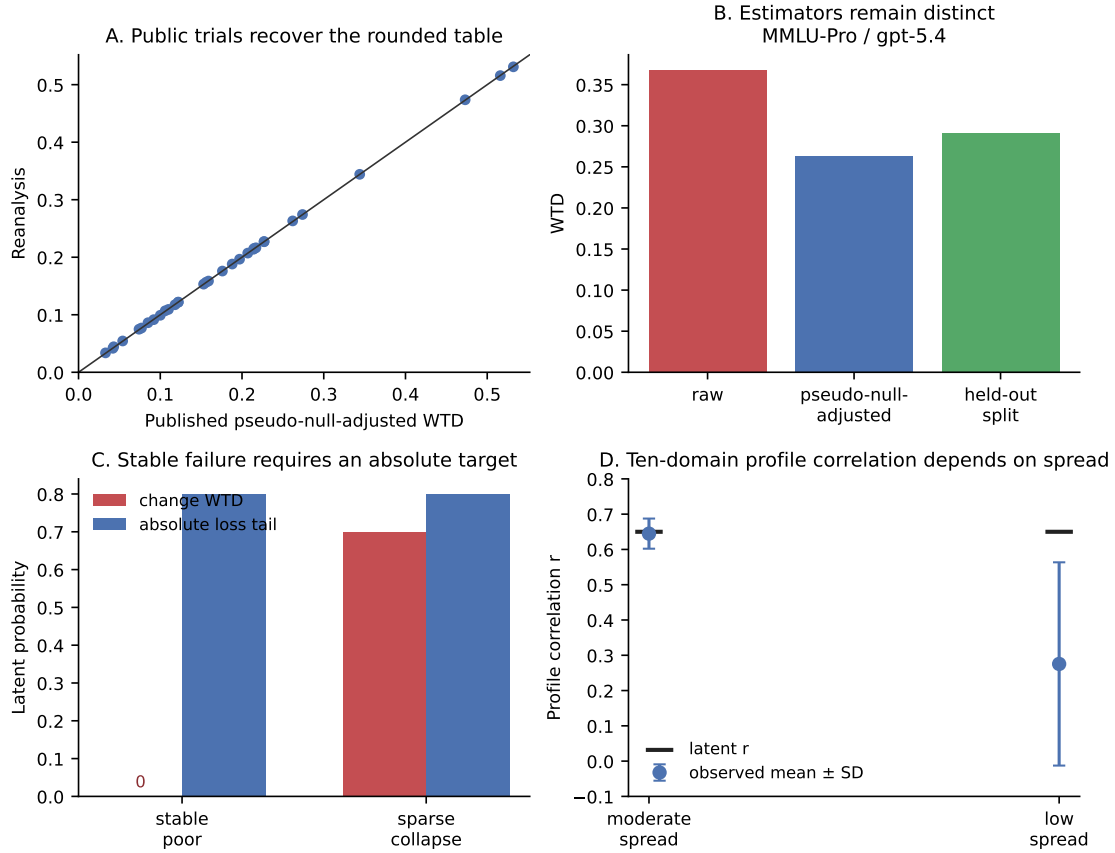


Figure 1: Released-output reproduction and known-truth checks. (A) Recomputed adjusted WTD for all 32 cells against the rounded published values. (B) Raw, pseudo-null-adjusted, and held-out split WTD for MMLU-Pro-gpt-5.4; the latter two estimate different estimands. (C) A stable-poor simulation has zero latent WTD but .80 absolute worst-decile loss. (D) With ten domains and latent $r = .65$, correlation estimation remains accurate at moderate profile spread but attenuates under low spread. Full settings and numerical outputs are in the empirical companion.

4.1 RELEASED-TRIAL REANALYSIS

The released design crosses four benchmarks with eight models and supplies 20 baseline and 20 context trials for every item in each of the resulting 32 cells. The reanalysis uses those responses rather than attempting to recreate outputs from proprietary model versions. It pins arXiv version 2, the source-repository commit, all 64 dataset revisions and file hashes, seeds, and analysis settings.

Recomputed values agree closely with the rounded table in the preprint: the largest absolute discrepancies are 0.053 percentage points for baseline accuracy, 0.20 points for signed change, 0.056 points for adjusted INS, and 0.135 points for adjusted WTD. The agreement checks the pipeline; it isn't independent validation of the estimands.

The MMLU-Pro-gpt-5.4 cell illustrates what the aggregate can't distinguish. Baseline accuracy is .8119 and context accuracy .7904, for a signed change of -.0215. Raw INS is .0673; subtracting the baseline pseudo-null contribution of .0212 gives .0461. Raw worst-decile WTD is .3680; subtracting its .1049 pseudo-null contribution gives an adjusted value of .2631. The held-out split estimate is

.2911. Thus a two-point average decline coexists with a .2631 all-trial adjusted WTD and a .2911 held-out split WTD. These aren't endpoints of an uncertainty interval.

The pseudo-null-adjusted and split-tail estimates shouldn't be treated as alternative implementations of the same correction. Split WTD exceeds pseudo-null-adjusted WTD in 31 of the 32 released cells, by .0331 on average, with differences from -.0012 to .0864. The pseudo-null procedure subtracts the expected response-noise contribution from a statistic selected and measured on all trials. The split procedure instead selects items on one response half and measures their changes on the other. The split procedure estimates the latent effect of a noisy-selected set, which needn't be the oracle tail defined by latent item effects.

4.2 KNOWN-TRUTH CHECKS

Simulation makes the target distinction observable. Under a null effect, mean adjusted INS, adjusted WTD, and split WTD are .0033, .0039, and -.0018 against a truth of zero. Depending on the estimator, 31–56% of the estimates are negative. Keeping those values untruncated is necessary if zero-centred correction is intended.

In a sparse-collapse scenario, split WTD averages .4514 against a mean truth of .4550 for the sets selected on noisy response halves, a bias of -.0036. The oracle worst-decile effect is .5500. Cross-fitting nearly removes reuse bias for its selected-set estimand without recovering the different oracle estimand. A report should state which tail it estimates rather than describe cross-fitting as a general debiasing step.

In the stable-poor scenario, the same low-performing items have unchanged response probabilities across conditions, so latent WTD is zero. Their absolute worst-decile loss is .80. The corresponding cross-fitted estimates are .0013 and .7911. A change tail and an absolute loss tail can disagree even without sampling error; the disagreement reflects their definitions.

A final ten-domain simulation probes profile correlation. With latent $r = .65$, moderate between-domain spread yields a mean observed $r = .645$ and root-mean-square error .043. Holding the latent correlation fixed while reducing profile spread lowers the mean observed correlation to .275 and raises root-mean-square error to .472. Nested-bootstrap coverage falls from .988 to .579 even as mean interval width increases from .249 to 1.111. When the perturbed latent profile is flat, its true correlation is undefined, but finite samples produce correlations with standard deviation .324. Raw profiles and their spreads are part of the evidence, not optional decoration around r .

The demonstration is deliberately bounded. It warrants fixed-data claims about the 32 released cells and conditional claims about estimator behaviour under the specified simulations. It can't establish prediction of deployment incidents, transport to a CHC battery, measurement invariance, retention, or component dependence. It illustrates the descriptive part of the framework; it doesn't supply the external outcome needed to validate projectibility.

5 FROM DESCRIPTIONS TO WARRANTED INFERENCES

The demonstration preserves distinctions that may be needed for inference or decision about a target. It doesn't establish which distinctions matter beyond the released cells or simulation designs. Messick's point returns here: predictive performance is one source of evidence for a proposed interpretation or use, not a complete theory of validity (Messick, 1995). Construct representation, response processes, comparability, consequences, and relevant external relations are additional, interacting considerations in the validity argument.

5.1 MATCH THE STATISTICAL DESIGN TO THE PROJECTION

There’s no single observational row for the framework. An item-generalization study may treat task-template clusters as the outer units. A perturbation-transport study may compare entire held-out families within systems. A model-population study needs enough independent lineages to hold out lineages rather than outputs. A temporal study requires a chronological split. Response replicates estimate cells inside these designs; they don’t multiply the number of independent systems, items, or contexts.

Let u index the independent unit appropriate to a declared projection, Y_u the external outcome, and X_u a preregistered set of baseline and granular predictors constructed without Y_u . Compare a simple model h_0 , such as equal-domain baseline performance or a prior incident rate, with a richer model $h_1(X_u)$. Under the target loss $\mathcal{L}_T$, the relevant increment is

$$\Delta_T = E_{\text{holdout}}[\mathcal{L}_T\{Y, h_0\} - \mathcal{L}_T\{Y, h_1(X)\}].$$

A positive estimate with decision-relevant uncertainty supports added predictive value within that holdout population. It doesn’t establish transport to another one.

The fitted model should reflect the design. Shared baseline observations induce dependence across perturbation cells; related versions share lineage; and the granular predictors are themselves estimated with error. Regularization or dimension reduction may be needed when many cells predict outcomes for few independent systems. Repeated-measures, hierarchical, or errors-in-variables models are often defensible, but no one form is mandatory. The paper’s requirement is that the row, dependence model, predictor construction, and outer holdout be stated.

For the running release decision, a random split of familiar items would test only item prediction. The stronger target requires development on earlier versions and context families, followed by assessment on a later version crossed with an untouched family. Thresholds, tail fraction, feature selection, and model tuning belong inside the development partition.

If the richer model exceeds a preregistered minimum improvement in target loss and calibrates the absolute tail constraint within tolerance there, it supports next-cycle prediction for that lineage and intervention range. Failure would restrict the report to the studied versions and context families rather than support the claimed next-cycle prediction.

5.2 SCORES AND PREDICTIVE MODELS IMPOSE DIFFERENT CONSTRAINTS

An interpretable score and a predictive model impose different constraints. A score such as

$$S_T = \sum_{g=1}^G w_g^{(T)} a_{g0}, \quad w_g^{(T)} \geq 0, \quad \sum_g w_g^{(T)} = 1,$$

retains an ordinary direction: more of each component can’t lower the total. A predictive or decision model may legitimately use negative conditional coefficients, interactions, nonlinearities, or a noncompensatory tail constraint. Convex weights shouldn’t be presented as the generally optimal predictive form.

Equal-domain weighting is a useful baseline because it’s transparent and difficult to tune. It isn’t measurement-neutral. Mapping heterogeneous domain scores to $[0, 1]$ doesn’t show that equal numerical differences have equal meaning. Alternative weights may be estimated with regularization, preregistered from an independent theory, or informed by a shrinkage prior. In every case, their value lies in held-out calibration, predictive loss, or decision value relative to the simple baseline.

5.3 CHC AND COMPARABILITY

CHC offers a developed vocabulary for broad human cognitive abilities (Carroll, 1993; McGrew, 2009; Schneider & McGrew, 2018). It can organize a broad AI battery without becoming an ontology of machine intelligence or a default machine weighting. Human loadings and the positive manifold don't automatically transport to artificial systems.

Profile comparison already assumes that domains and scoring rules retain sufficiently similar meaning across systems. Rubric-invariance checks, family-specific item analyses, differential-item-functioning analyses, or multi-group item-response models can test that assumption when the data support them (Meredith, 1993). This paper doesn't conduct a full invariance study. It treats CHC as a defeasible measurement vocabulary: compare it with simpler and alternative decompositions, and retain it only within the targets and populations for which comparability and external performance hold.

5.4 DECISION USE

Decision uses require losses and constraints at the start, not only after the descriptive statistics have been calculated (Keeney & Raiffa, 1993). False reassurance, false alarms, and severe tail events can have different costs. A release rule may combine a calibrated expected-loss model with a noncompensatory WTL constraint. When WTL is declared noncompensatory, a stable mean can't override its violation; an imprecise tail estimate still can't support a confident action.

No generic capability–stability tiers follow from this framework. Thresholds for monitoring, release, or independent review have to be calibrated against relevant outcomes. Reports should also keep base-model and deployed-system claims separate and identify every untested transport step.

5.5 BOUNDED EMPIRICAL CLAIMS

The proposal entails comparative claims that should be preregistered within a bounded study universe.

P1: Some granular descriptions justify their measurement cost for some targets. Before a study, name the target class, system population, minimum useful reduction in held-out loss or calibration error, and sampling cost. A component fails for that class if it doesn't meet the threshold with adequate precision. It should then be removed from that target-specific model, even if it remains descriptively interesting.

P2: Useful granularity varies by target. Register a finite set of targets and compare the best regularized model built from granular descriptions with simple aggregate baselines under a common outer validation design. If the simple model remains within a preregistered equivalence margin across that universe, the added granularity doesn't improve enough to justify inclusion there.

P3: Weights and models have limited transport unless demonstrated otherwise. Register source and target populations and a maximum acceptable loss of calibration or decision value. If one score or model repeatedly transports within that tolerance across target changes, a more general scheme becomes empirically defensible over that range.

These claims don't predict that every granular description will help or that transport will fail. Their falsifiable content lies in the registered target universe, comparison, margin, and cost. Predicting a mechanistic label from behaviour may make the descriptive set a useful proxy within a population; it doesn't remove the logical underdetermination of mechanism by behaviour.

6 APPLICATIONS OF THE PROJECTION DECLARATION

Retention, feedback response, and component dependence aren't constitutive parts of one robustness construct. They're examples of how the declaration changes when the time horizon, intervention, or bearer changes.

6.1 RETENTION OF ABOVE-PRETEST GAIN

For a learning claim, test performance before instruction, immediately afterward, and at each declared clean-session delay. Let these item-level scores be b_t , u_t , and v_{td} . Define above-pretest gains

$$g_{t0} = (u_t - b_t)_+, \quad g_{td} = (v_{td} - b_t)_+,$$

and total immediate above-pretest gain $M_0 = \sum_t g_{t0}$. Define

$$R_d^{\text{gain}} = \frac{\sum_t \min(g_{t0}, g_{td})}{M_0} \quad (4)$$

With a preregistered minimum average gain $\gamma_{\min} > 0$, interpret this ratio only when $M_0/N \geq \gamma_{\min}$. Report the pretest, immediate score, delayed score, denominator, threshold, and signed item changes. A high ratio with negligible immediate gain isn't evidence of durable learning.

Interpreting above-pretest gain as caused by instruction requires additional design support. Novel arbitrary associations can reduce prior-knowledge concerns, while a matched no-instruction group can address repeated testing or other rival causes. Independent screening and estimation samples reduce regression to the mean. Delays should remain separate unless a declared temporal target supplies weights.

6.2 RESPONSE TO FEEDBACK

Let $e_k \in [0, 1]$ be error at attempt $k = 1, \dots, K$. Define cumulative decreases, cumulative increases, endpoint gain, and total variation:

$$D = \sum_{k=2}^K (e_{k-1} - e_k)_+, \quad U = \sum_{k=2}^K (e_k - e_{k-1})_+,$$

$$G = e_1 - e_K = D - U, \quad V = D + U.$$

Report G and U as primary trajectory summaries, with $U/(K-1)$ when attempt counts differ. The retired backsliding share satisfies

$$B = \frac{U}{V} = \frac{V - G}{2V} \quad (V > 0).$$

Because B is determined by G and V , it adds no independent output and has no continuous extension at a flat trajectory.

Repeated attempts can improve performance without feedback through resampling, extra computation, or task exposure. A feedback claim requires a matched no-feedback control with attempt count, token budget, tools, timing, generation settings, and scoring held constant. Report ΔG and ΔU between conditions with uncertainty at the randomization unit. "Try again", self-review, tool output, and explicit error diagnosis are different interventions.

6.3 BEARER AND COMPONENT DEPENDENCE

A base model and a deployed model–tool–retrieval composite are different bearers of a score. For a declared composite, compare the intended configuration, selected component degradations, and a minimal model-only configuration when that contrast is decision-relevant. Tools and retrieval may interact, so a small factorial design can be more informative than degrading one component at a time.

Level, item change, WTD, and absolute WTL should be reported for every transition. A gradual performance change across the tested degradation levels supports inference only about those levels; a large decrease after altering a component indicates dependence on it. Neither result identifies the “real” agent.

Construct and mechanistic claims require additional evidence. A multimodal system can score well on visual questions because its responses can depend on language cues even without image input (Asadi et al., 2026). Behavioural testing alone also underdetermines whether performance relies on a representation; probes and causal interventions can strengthen that claim (Arora et al., 2026; Yetman, 2026). These provide additional validity evidence; they don’t follow from stability in the descriptive report.

7 WORKFLOW AND LIMITS

7.1 OPERATIONAL SEQUENCE

The design can be implemented without access to model internals:

1. Register the projection declaration. Name the claim, system and task populations, bearer, intervention range, horizon, decision criterion, and matching evidential test.
2. Audit the sampling facets. Record systems and lineages, item and template clusters, perturbation families and realizations, response replicates, scorers, and time; mark each fixed, sampled, crossed, or nested.
3. Fix the measurement design. Specify paired items, rubrics, target loss, tail fraction and minimum count, perturbations, controls, sampling budget, and the distinction between fixed-set, stochastic-response, and population inference.
4. Run balanced conditions. Randomize order where possible, retain item identifiers, and prevent data used for adaptation, tail selection, or threshold setting from leaking into final estimation.
5. Publish the descriptive report. Include profiles, r , L , INS, directional components, WTD, baseline and perturbed WTL, raw and adjusted estimates, and cellwise or simultaneous uncertainty.
6. Validate the intended projection. Compare a simple baseline with the richer score or model on the outer holdout named in advance. Report calibration, target loss, and uncertainty in the incremental value.
7. Release an audit trail. Publish code, data provenance, seeds after testing, model and scorer versions, exclusions, missingness, compute use, split assignments, and preregistered deviations.

Lottery-style sampling from declared perturbation families can reduce direct optimization against known test cases. It doesn’t make the sample representative or the result intrinsically valid. The sampled conditions still have to support the intervention range in the declaration.

For the running release decision, the final report would identify the later system version and untouched context family, show absolute WTL beside change-based WTD, and evaluate the release rule under the declared loss. If the outer family was used to choose q , features, or thresholds, the

advertised projection remains untested.

7.2 LIMITATIONS

First, disaggregation doesn't repair construct-invalid tasks, ambiguous scoring, or poor coverage. Nor does this revision establish measurement invariance for CHC domains across artificial systems.

Second, repeated sampling and tail estimation can be expensive. Adaptive allocation may reduce cost but creates additional selection and dependence that the analysis should represent. Deterministic fixed-set reports avoid response-repetition cost while remaining silent about wider item or context populations.

Third, a perturbation is informative only relative to a property expected to remain invariant. Task-irrelevant context is designed to preserve the answer; distribution shifts, tool changes, and component removal may change the task itself. The evaluator has to defend the comparison.

Fourth, tail membership and item effects may not transfer across models or versions. Zhang et al.'s model-specific changes are evidence against assuming a common risk set, not evidence that no transport is possible.

Fifth, the empirical demonstration in Section 4 contains no external deployment outcome and doesn't validate a general predictive composite.

Finally, the framework neither defines AGI nor settles whether a system represents, understands, consolidates, or controls. It asks what a finite evaluation supports and what additional evidence a broader claim would require. That narrower question applies beyond AGI, even though AGI claims make it especially pressing.

8 CONCLUSION

The same aggregate accuracy can describe unchanged items, balanced improvements and degradations, or stable severe failures. A domain profile repeats that compression at a coarser level.

The response has three moves: declare the projection target and scope; preserve domain-profile correlation, signed level, item instability, directional components, worst-tail degradation, and absolute worst-tail loss; then test the projection with a holdout on the corresponding facet. Held-out items, perturbation families, system-dependency groups, and later releases answer different questions.

The demonstration warrants only fixed-data and conditional simulation claims, not transport to deployment. Projectibility names the additional target-indexed support needed for inference beyond the observations used to construct a score. It's part of validity, not a property of a descriptive report. The discipline is general: declare, preserve resolution, and test with evidence drawn from the corresponding scope. Only then is aggregation warranted for that target.

A FORMAL SANITY CHECK OF THE ESTIMANDS

A companion Lean 4 module proves raw-estimand bounds and the identities

$$|L| \leq \text{INS}, \quad L = F^+ - F^-, \quad \text{INS} = F^+ + F^-,$$

It also verifies $G = D - U$, $V = D + U$, and $|G| \leq V$.

Tail membership remains external; the code covers neither noisy-tail identification, adjustment, cross-fitting, correlation, nor sampling. Adjusted estimates can violate raw bounds. These results establish neither reliability nor projectibility.

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